

RANKUNO AI ENGINE PLATFORM SPECIFICATION

Phase 1: Page Classification & Intent Analysis Engine

Master Technical Architecture, 6-Signal Consensus Pipeline, Graph Site Crawler ($G = (V, E)$), & Security Blueprint

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Target Coverage: **$\geq 98\%$ Accuracy**

Status: **Binding Specification**

1. Executive Summary & Problem Statement

Modern enterprise web applications (e.g. Macy's, Shopify Plus, B2B SaaS portals) possess non-linear, complex Information Architectures. Legacy SEO crawlers evaluate websites using rigid mechanical rules that fail on real-world production environments:

● The Flat URL Failure

Traditional tools rely on URL path slashes (`/shop/women/`). When encountering a **Flat URL** sitting directly off root (e.g., `macys.com/diamond-pendant-necklace`), legacy crawlers dump it into an unorganized "Root / Other" bucket.

● The Click-Depth Fallacy

Screaming Frog calculates depth based on HTTP click distance from the homepage. A blog post linked on the homepage is assigned **Depth 1**, misclassifying a leaf page at the same level as major category hubs.

2. Competitive Audit: Legacy Tools vs. Rankuno AI Engine

Capability / Metric	Screaming Frog v20+	Semrush (2026)	Botify / Lumar	Rankuno AI Engine
Flat URL Resolution	FAILS (ROOT BUCKET)	FAILS (PATH-BASED)	BASIC REGEX	100% (6-SIGNAL CONSENSUS)
Hierarchical IA Graphing	CLICK DEPTH GRAPH	FOLDER LIST	MANUAL RULES	DECOUPLED L0-L3 GRAPH
Speed for 20,000 Pages	45–90 Mins	30–60 Mins	10–20 Mins	15–30 SECONDS (CASCADING)
API Cost per 20k Scan	\$0 (Local Desktop)	Subscription	Enterprise Plan	~\$0.10 – \$0.50 (95%+ \$0 LOCAL)
Automated Remediation	NO (STATIC CSV)	NO (STATIC CSV)	NO (STATIC CSV)	AUTO-GENERATES CODE / PATCHES

3. The 4-Layer Cascading Pipeline Architecture

To process 20,000+ pages in under 30 seconds with near-zero API cost, Rankuno uses a 4-Layer Cascading Pipeline:



4. The 6-Signal Consensus Engine & Decoupled Taxonomy

Rankuno decouples **Hierarchy Level (L0, L1, L2, L3, Utility)** from **Primary Page Type (Product, Service, Article, Lead Gen)**:

Structural Hierarchy (`HierarchyLevel`)

- `L0_HOMEPAGE` : Root Entry Point
- `L1_PRIMARY_NAV_HUB` : Primary Section (Services, Products)
- `L2_SUB_NAV_HUB` : Sub-Category / Collection
- `L3_LEAF_PAGE` : Product SKU, Service Page, Article

The 6 Consensus Signals

- Signal 1:** ARIA Navigation Tree Analysis
- Signal 2:** CMS API Endpoint Footprints
- Signal 3:** XML Sitemap Index Parsing
- Signal 4:** Schema.org `@graph` JSON-LD
- Signal 5:** Internal Link In-Degree Centrality
- Signal 6:** LLM Zero-Shot Fallback

- **UTILITY_PAGE** : Terms, Privacy, Faceted Filter

5. Pydantic Data Contracts

```
class SiteNodeIntelligence(BaseModel):
    url: str
    canonical_url: str
    depth_from_l0: int
    hierarchy_level: HierarchyLevel
    primary_page_type: PrimaryPageType
    topical_cluster: str
    parent_node_url: Optional[str]
    breadcrumb_path: List[str]
    is_cross_silo_link: bool
    inbound_internal_links_count: int
    outbound_internal_links_count: int
    confidence_score: float = Field(ge=0.0, le=1.0)
    classification_method: str
    reasoning: str
```

6. Security Hardening & Vulnerability Matrix

Threat / Vulnerability	Root Cause	Rankuno Mitigation Guardrail
SSRF (Server-Side Request Forgery)	Malicious target URL pointing to internal IPs (127.0.0.1 , 169.254.169.254).	IP Whitelist & Private Range Blocker: Rejects RFC1918 private IPs before network socket connection.
Prompt Injection via Web Content	Hidden text in scraped HTML attempting to override LLM instructions.	Input Sanitizer & Pydantic Enforcement: Strips hidden CSS/script blocks and enforces Pydantic StrictModel schema validation.
Unbounded Memory Leaks (20k URLs)	Accumulating 20,000 raw DOM payloads in Python memory.	Streaming Batch Pipeline: Processes in 500-URL chunks, flushes to DB/Disk, and triggers explicit GC.

7. Open-Source Models & Platform Leverage

- HTTP Scraper:** httpx + selectolax + trafilatura (\$<5\text{ms}\$ DOM parsing).
- Headless Fallback:** Playwright (Dynamic JS rendering for React/Next.js SPAs).
- Local Zero-Shot ML:** MoritzLaurer/DeBERTa-v3-large-mnli (Hugging Face).
- Open-Source LLM:** Qwen/Qwen2.5-14B-Instruct (Hugging Face / Ollama).
- Cloud LLM API:** Google Gemini 2.0 Flash (Bulk fallback).